Zhiheng Weng

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Research Interests

Human-computer interaction and learning sciences. I am interested in how adaptive guidance and AI support **help older adults learn, retain, and transfer skills** for independent use of everyday and health technologies.

Education

University of Illinois Urbana-Champaign Aug 2024 - Expected Dec 2026

B.S. in Computer Science + Education (Learning Sciences) GPA 3.76/4.00

James Scholar Honors and Dean's List (Fall 2025 - Present).

Research Experience

Department of Educational Psychology, UIUC Spring 2026

Undergraduate Researcher | College Education and Social-Emotional Development

- Contributed to a study examining the effects of college education on social, emotional, and behavioral skills using a longitudinal dataset of approximately 30,000 people.
- Supported research design, data organization, and pipeline setup for Causal Forest modeling within an augmented inverse probability weighting (AIPW) framework.

Undergraduate Research Scientific Advancement, UIUC Spring 2026

Undergraduate Researcher | LLM-Based Mathematical Equation Relation Extraction

- Completed 100+ annotations for a mathematical derivation graph dataset supporting LLM-based equation relation extraction.
- Contributed to annotation methodology and dataset quality assurance.

Learning Design and Teaching

Question Everything 2026 - Present

Designer and Developer | Independent project | huitiwen.cn

- Built a 24-lesson bilingual curriculum and a self-paced website with 12 interactive missions for children ages 6 to 11.
- Built activities that record learners' reasoning before feedback and preserve revisions. Added optional hints, unfamiliar assessment cases, and scoring criteria.
- Interviewed an elementary educator and reduced a question set from 10 to 4 in response to concerns about text and information load.
- Prepared pre/post assessment and pilot materials. Learner testing is pending.

Siebel School of Computing and Data Science, UIUC Aug 2025 - Aug 2026

Course Assistant | CS124 Introduction to Computer Science

- Supported a 300-student introductory programming course, guiding problem decomposition across programming, data structures, and algorithms.
- Used weekly office hours to identify conceptual gaps and explain technical ideas to students with different learning backgrounds.

Publication

Weng, Z. (2023). Prediction and Recognition of Epileptic Seizures Based on Artificial Intelligence. *Highlights in Science, Engineering and Technology*, *72*, 1092-1098. doi:10.54097/gp3axg67

Health Technology and AI Projects

OhMyTSS Nov 2025 - Present

Co-Founder and Developer | Swift and Apple HealthKit

- Integrated HealthKit workout data into an iOS application that presents readiness and training-load insights, with data processing on the device.
- Investigated user-reported issues, corrected data parsing and validation, and shipped stability improvements for active users.

Seizure Prediction Wristband 2022 - 2024

Initiator and Team Lead | C/C++ and Wearable Sensors

- Developed a wearable research prototype, from hardware and sensor data collection to time-series features and a classification model evaluated through cross-validation.
- German utility model for an epilepsy early-warning bracelet, DE202023107669U1 (January 2024).

Meeting Minion Aug 2025 - Jan 2026

Team Lead and Developer | AI Meeting Transcription and Summarization

- Led a team building and deploying a meeting transcription and summarization application used by real users.
- Integrated WhisperX, local LLM inference through Ollama, and SQLite. Maintained CI/CD, pull-request reviews, and a pytest suite with 64+ tests.

Engineering Experience

Shenzhen Smart City Technology Development Group Summer 2026 - Present

Software Engineering Intern | Shenzhen, China

- Built a crowd stress/fatigue simulation model and debugging dashboard in C++ and Unreal Engine 5. Extended navigation with doorway flow limits and queue-aware exit costs.
- Implemented fixed-timestep regression runs and SHA-256 checks across 57,754 trajectory rows, reducing each verification round from approximately 40 to 5.5 minutes.

Technical and Research Skills

Programming · Python, C/C++, Swift, JavaScript, HTML/CSS

Tools and systems · HealthKit, Unreal Engine 5, Django, SQLite, WhisperX, Ollama, pytest, CI/CD

Research and design · Data annotation and quality assurance, longitudinal data preparation, learning task and assessment design, educator interviewing

Service and Honors

Undergraduate Research Scientific Advancement, UIUC Fall 2026 - Present

Webmaster and Secretary. Maintain the website and respond to inquiries.

USACO Gold Division (2021-2022) · AMC 12 top 1%, 135/150 (2022) · AIME 11/15 (2023)

Additional Training

MIT Game Lab Summer School (Summer 2023) · Stanford AI4ALL (Summer 2020)